

**FORCES EXTRACTION FOR ANALYSIS OF
BOLTED JOINTS**

Section: **3.2 – MECHANICALLY FASTENED JOINTS**

Chapter: **3215**

Title: **FORCES EXTRACTION FOR ANALYSIS OF BOLTED JOINTS**

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1 INTRODUCTION

This chapter provides guidance on how internal loads (either available from global finite elements models (GFEM) or other means) can be derived into the specific loads sets necessary for calculating each type of bolting group in an airframe.

Different idealization of the structural elements in the GFEM shall impose different ways of extracting the necessary input loads for later bolting groups analyses (e.g. 1D or 2D idealization of the frames of a fuselage, explicit differentiation of an interface element for a hinged joint, etc...). This chapter provides an overview/summary of all the types of fastened connections in the airframe. Further and detailed description of these methodologies for particular airframe parts will be given in the assembly methods chapters.

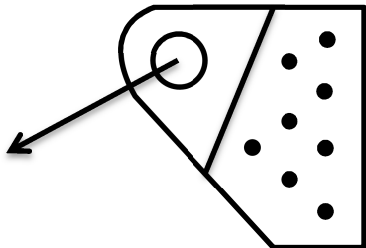
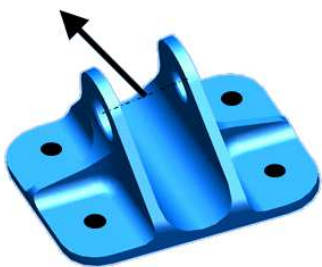
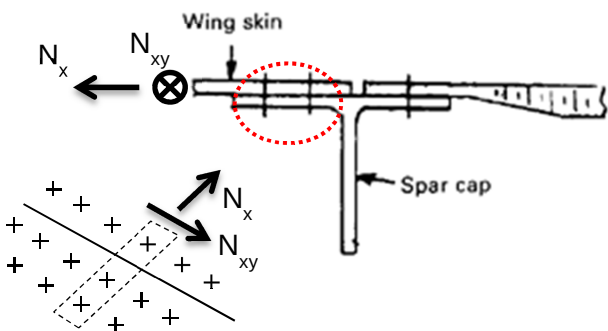
This chapter will be completed with more standard types of fastened connections.

2 BOLTING GROUPS LOADS

Tables in following pages show the common types of bolting groups and associated design details in an airframe, explain how to process and reduce airframe internal loads to a valid input set for the rivets loads calculation, and points to the most appropriate methodology for that task.

Further guidance of the selection of the most appropriate method for calculating and sizing the fastened attachments of common airframe components can be found in chapters of sections 3.3 "METAL ASSEMBLIES" and 3.8 "COMPOSITE ASSEMBLIES".

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BOLTING GROUP	METHOD
	Fitting Plane Joint Method: See chapter §3213 (alternatively Bruhn D.1.17 or Niu 9.4) Tool: STRENGTH2000 – Module NIE3/BOLTS2D Loads: Force at interface point (in-plane)
	Fitting Plane Joint with out-of-plane loads Method: See chapter §3213 Tool: STRENGTH2000 – Module TBD Loads: Force at interface point
	Fitting Joint (not plane) Method: See chapter §3214 Tool: STRENGTH2000 – Module BOLTS3D Loads: Force at interface point
	Lap Plane Joint Continuous joint (quasi-1D analysis of the slice) Method: See chapter §3211 Tool: STRENGTH2000 – Module TBD Loads: In-plane loads (forces per unit length), axial (normal to riveting line) and shear (*)

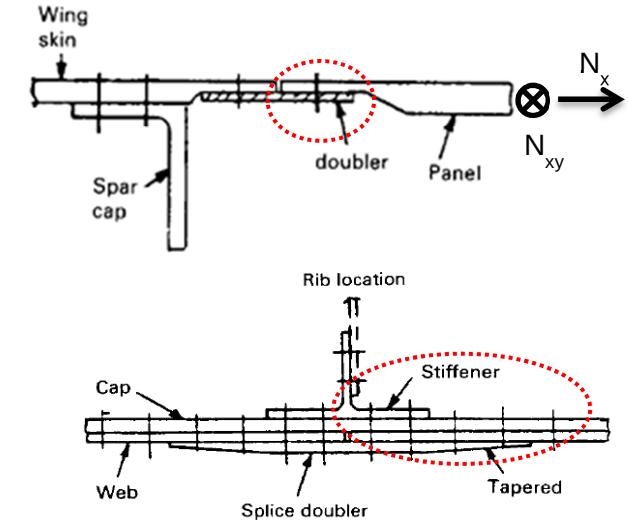
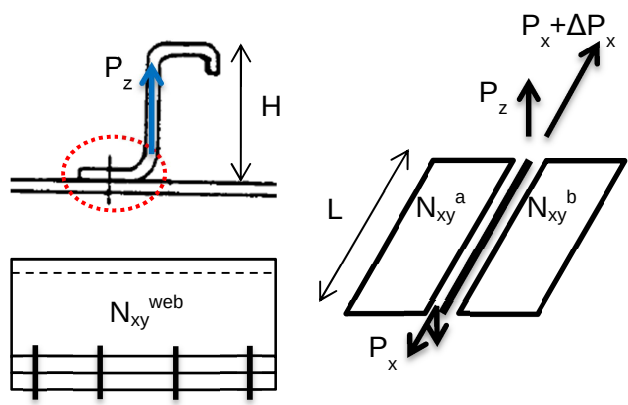
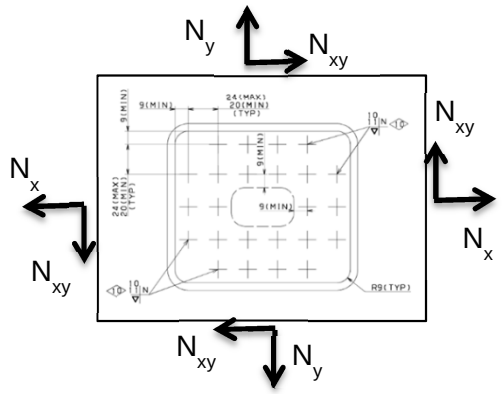
(*) By-pass axial load parallel to the riveting line may be relevant for the later strength check (inter-buckling or net-section failure) even not used for calculating rivets loads. Single lap shear configurations shall induce additional out plane effects that are normally neglected in calculations (implicitly accounted in allowables).

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DEFENCE AND SPACE

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BOLTING GROUP	METHOD
	Butt Plane Joint (with doubler/strap) Continuous joint (quasi-1D analysis of the slice) Method: See chapter §3211 Tool: STRENGTH2000 – Module TBD Loads: In-plane loads (forces per unit length), axial (normal to riveting line) and shear(*)
	Stringer/Frame/... to skin Continuous joint (quasi-1D analysis of the slice) Method: See chapter §3211 Tool: STRENGTH2000 – Module TBD Loads: Shear at the interface riveting line (**) - Delta shear between skin panels ($N_{xy}^a - N_{xy}^b$), or... - Shear in stringer web ($\approx P_z/H$ or N_{xy}^{web}), or... - Delta axial force along stringer ($\approx \Delta P_x/L$)
	Plane Joint (with significant 2D effects) Method: See chapter §3212 Tool: Typically a dedicated FEM is recommended. However, if design is appropriate (mitigates loads redistribution), an extension of the quasi-1D methods might be suitable Loads: Far-field/gross in-plane loads (forces per unit length), axial in both directions and shear

(*) By-pass axial load parallel to the riveting line may be relevant for the later strength check (inter-buckling or net-section failure) even not used for calculating rivets loads. Single lap shear configurations shall induce additional out plane effects that are normally neglected in calculations (implicitly accounted in allowables).

(**) Pressure on skin and/or curvature (frames) may cause additional tension/pull-out forces on these rivets

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3 SCOPE AND LIMITATIONS

Calculation of rivets/bolts loads in a mechanically fastened joint between two (or more) structural components.

See referred chapters for details on specific limitations of each of the methods proposed.

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
H	mm	Height (e.g. of a stringer)
L	mm	Length
N	N/mm	In-plane force per unit length in a panel, plate or shell
P	N	Force in a uni-dimensional element or at an interface point
TBC	-	To Be Confirmed
TBD	-	To Be Defined
X, Y, Z	-	Directions of a rectangular reference axis system

REFERENCES

N/A

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures
NASTRAN (MSC-Software)	2005	Multidisciplinary structural analysis software based in the Finite Elements method

Table 1. *Software programs referenced in this chapter*

FORCES EXTRACTION FOR ANALYSIS OF BOLTED JOINTS

DISTRIBUTION LIST

<i>Department</i>	<i>Name</i>	<i>Full Doc.</i>	<i>Cover Page</i>
Stress Methods & Optimisation (TEASP-TL3)	Fernass Daoud	X	
Fatigue & Damage Tolerance GET (TEASA-TL1)	Javier Gomez-Escalonilla Martin	X	
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
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Table 2. *Distribution list (of electronic document, ahead of public release)*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 01/08/2018	G. Baños	First issue	All pages

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Analysis (TEASA)	Miguel Ángel Calero 01/08/2018	Bernhard Mayer 01/08/2018
Stress Methods (TEASP-TL3)	G. Baños 01/08/2018	F. Glock 01/08/2018

Table 4. *Approval signatures table for last revision of this chapter*